Solution Design Document – YouTube-trending-Analyser

Version History

v1.0 Initial draft

Executive Summary

This project aims to build a YouTube Trending Analysis System that analyzes historical YouTube video data to identify patterns in video performance and trending behavior. The system uses data analytics and machine learning models to explore engagement metrics such as views, likes, comments, categories, and publishing time.

The goal is to transform raw YouTube datasets into actionable insights and predictive analytics that help understand what factors contribute to trending videos.

Purpose of Solution Design Document

This document defines the design, scope, architecture, and technical implementation of the YouTube Trending Analysis System.

It provides a structured description of:

- system objectives
- data processing workflow
- machine learning modeling approach
- deployment architecture

The document ensures alignment between business requirements, technical implementation, and project deliverables.

Project Background

YouTube generates massive amounts of content daily, and understanding what factors influence video popularity is valuable for content creators and analysts.

However, raw YouTube datasets are difficult to interpret without proper analysis.

This project addresses that problem by:

- analyzing historical YouTube trending data
- identifying engagement patterns
- Building machine learning models to predict video performance.

Project Team

Project Developer – Yashwanth Doddapaneni

Point of Contact

Project Manager – Instructor– Ruchik Kota

Business Requirements

The system should:

- analyze YouTube trending datasets
- identify patterns in engagement metrics
- visualize trends using dashboards
- provide predictive insights using machine learning models

The goal is to convert raw data into **interactive analytics and predictive insights**.

Scope

The project includes:

- data ingestion and preprocessing
- exploratory data analysis (EDA)
- machine learning modeling (Linear Regression initially)
- interactive dashboard development using Streamlit
- visualization of trends and insights
- Automate recommendations Using Gen AI

Success Criteria

The project will be successful if the system:

- successfully processes YouTube datasets
- identifies meaningful engagement patterns
- trains machine learning models to predict video performance
- provides interactive visual analytics through the dashboard

Solution Overview

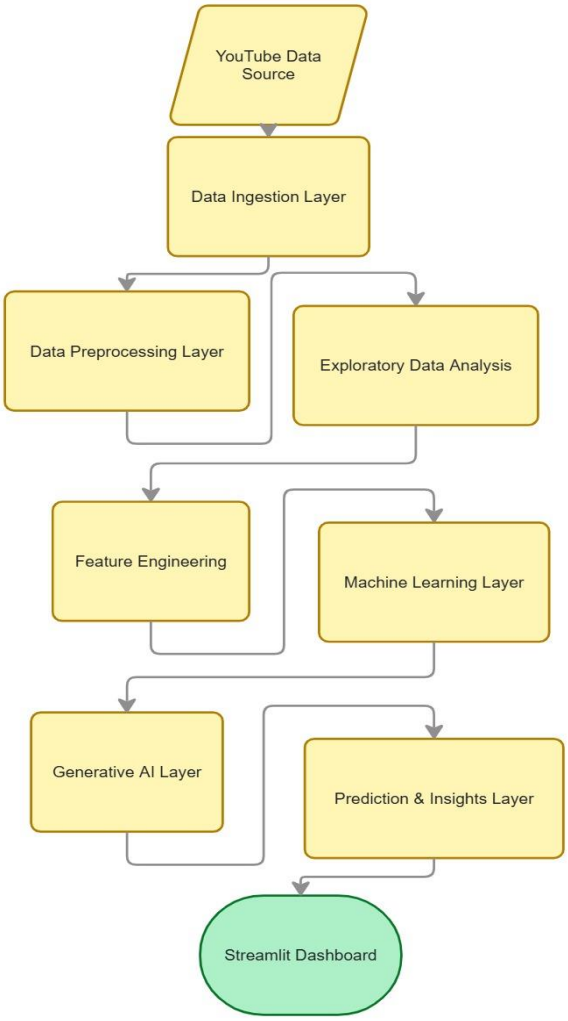
The solution consists of a **data analytics pipeline combined with machine learning models**.

Main components:

1. Data ingestion and preprocessing
2. Exploratory data analysis
3. Feature engineering
4. Machine learning modeling (Linear Regression)
5. Interactive dashboard visualization

The system will allow users to explore trends and analyze video performance metrics.

Automation Flow



Implementation Timeline

6 weeks total – 2 weeks data prep, 2 weeks modeling, 2 weeks deployment.

Technology Stack

Programming Language - Python

Data Processing - Pandas, NumPy

Visualization - Matplotlib, Seaborn

Machine Learning - Scikit-learn

Application Framework - Streamlit

Development Environment - Jupyter Notebook / VS Code

Version Control - GitHub

Current State Process Flow

Currently, the dataset exists as raw data files with no automated analysis or prediction system.

The current workflow requires manual data exploration and does not provide predictive insights or automated trend analysis.

Functional Requirements

The system currently should:

- load and process YouTube datasets
- generate visual analytics for engagement metrics
- allow users to explore trends by category or metrics
- train machine learning models to predict video performance
- display insights through an interactive dashboard

Technical Requirements

- ability to process large datasets efficiently

.

Assumptions and Considerations

- the dataset contains sufficient historical YouTube data

- engagement metrics such as views, likes, and comments are available
- data quality is sufficient for meaningful analysis

Risks and Dependencies

Potential risks include:

- incomplete or inconsistent data
- limitations in dataset size or features
- model performance depending on data quality

Dependencies include:

- Python libraries
- dataset availability
- computing resources for model training